NOTICE OF CHANGE

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MIL-STD-2082A(AS)
NOTICE 1
14 September 1990

MILITARY STANDARD

ASSEMBLY PROCEDURES, GYRO

TO ALL HOLDERS OF MIL-STD-2082A(AS):

1. THE FOLLOWING PAGES OF MIL-STD-2082A(AS) HAVE BEEN REVISED AND SUPERSEDE THE PAGES LISTED:

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
iii	26 Feb 1985	iii	REPRINTED WITHOUT CHANGE
iv		iv	26 Feb 1985
1		1	26 Feb 1985
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3		3	26 Feb 1985
4		4	26 Feb 1985
5		5	26 Feb 1985
6		6	26 Feb 1985
9	26 Feb 1985	9	REPRINTED WITHOUT CHANGE
10		10	26 Feb 1985
23		23	26 Feb 1985
24	26 Feb 1985	24	REPRINTED WITHOUT CHANGE
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32		32	26 Feb 1985

2. RETAIN THIS NOTICE AND INSERT BEFORE TABLE OF CONTENTS.

3. Holders of MIL-STD-2082A(AS) will verify that page changes and additions indicated above have been entered. This notice page will be retained as a check sheet. This issuance, together with appended pages, is a separate publication. Each notice is to be retained by stocking points until the military standard is completely revised or canceled.

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FOREWORD

This standard has been prepared to define inspection and assembly procedures for Sidewinder Gyro Assemblies and is suitable for citation in request for proposals, invitation for bids, contracts and purchase orders, or governing specification.

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1. SCOPE

1.1 AI Scope. This standard describes assembly, machining, and inspection procedures for Sidewinder gyro assemblies. The gyro assembly is an integral unit of the guidance and control section, WGU-4/B and WGU-4A/B.

1.2 AI Paragraph coding. Each paragraph and table is coded as follows:

M meaning Mandatory; i. e., use instruction as stated, except for tooling.

MXM meaning Mandatory Except for Method.

AI meaning Advisory Information.

OP meaning Optional

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

A-A-208	Ink, Marking Stencil, Opaque (Porous and Nonporous Surfaces).
O-C-265	Chemicals, Analytical; General Specification for.
O-E-760	Ethyl Alcohol (Ethanol); Denatured Alcohol; and Proprietary Solvent.

MILITARY

MIL-S-22473	Sealing, Locking and Retaining Compounds: (Single-Component).
MIL-A-46050	Adhesives, Cyanoacrylate, Rapid Room-Temperature Curing, Solventless.
MIL-G-85105	Gyro Assembly, AIM-9M (Sidewinder).

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STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-2081 Assembly Procedures, Gimbal.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (Cage Code 30003)

2104343 Lacquer, Special Optical Black.

2251012 Balance Weight, Optical Balance.

2412395 Sheet, Technical, Frequency, Specifications.

639AS516 Nut, Sunshade.

639AS534 Weight, Dynamic Balance.

639AS537 Post, Gimbal.

639AS572 Screw, Balance.

639AS577 Shim, Focusing.

639AS622 Sunshade-Damper Assembly.

639AS1467 Nut, Preload.

639AS1486 Adhesive, Epoxy-Polyamide.

639AS1487 Adhesive, Epoxy.

639AS1574 Adhesive, Methacrylate, Solvent Solution.

639AS1611 Cement, Liquid Cellulose Nitrate.

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DRAWINGS - Continued

NAVAL AIR SYSTEMS COMMAND (Cage Code 30003)

639AS3010	Head Coil Assembly.
639AS3100	Seeker Section, Infrared, Guided Missile.
639AS3112	Mirror, Primary.
639AS3122	Aft Housing and Lens Assembly.
639AS3142	Guard, Edge.
639AS3164	Mirror, Secondary, One-piece.
639AS3169	Gimbal Assembly.
639AS3178	Gyro Subassembly.
639AS3180	Gyro Assembly.
639AS3181	Screw, Socket Body.
639AS3182	Stud, Mirror.
639AS3184	Optical Barrel, Machined.
639AS3193	Mirror, Primary, Replica.
639AS3962	Seeker Section, Infrared, Guided Missile.
639AS6502	Gyro Assembly.
639AS6503	Gyro Subassembly.

(Unless otherwise indicated, copies of Naval Air Systems Command drawings are available from the Naval Air Technical Services Facility (NATSF) (Code 312), 700 Robbins Avenue, Philadelphia, PA 19111-5097.)

PUBLICATIONS

NAVAL AIR SYSTEMS COMMAND (Cage Code 30003)

AD-874	Clean Room Conditions.
AD-883	Gyro Specialist and Gyro Worker Qualifications.

(Copies are available from the Naval Air Systems Command, Standardization Section (Code 51122E), Washington, DC 20361-5110.)

Supersedes page 3 of 26 February 1985

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2.2 Non-Government publications. The following documents form a part of this document to the extent satisfied herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

AFBMA Standards Annular Bearings Engineers Committee of the Anti-Friction Bearing Manufacturers Association (AFBMA) (Twelve Sections).

(Application for copies of AFBMA Standards should be addressed to the Anti-Friction Bearing Manufacturer's Association, 60 East 42nd Street, Rm 2825, New York, New York 10017.)

"Adhesive Bonding Techniques and Applications" by Charles V. Cagle, McGraw-Hill, 1968.

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS (Not Applicable)

4. GENERAL STATEMENT OF REQUIREMENTS

4.1 M End product. The end product gyro assembly shall comply with all listed applicable documents set forth in section 2 of this standard and as specified herein.

4.1.1 M Production techniques. Production fabrication and assembly techniques are controlled by process sheets for all machining, inspection, and assembly phases. The instructions herein are intended to ensure acceptable quality products.

4.1.2 M Process documents. All manufacturing and inspection techniques and procedures shall be documented and made available to the particular work station involved. These procedures shall conform to the basic principles of inspection, machining, and assembly as set forth in this standard.

4.1.3 M Clean room. Clean room assembly conditions shall be in accordance with AD-874. Assembly personnel shall be qualified in accordance with AD-883.

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4.2 MXM Equipment. This section lists the commercial products and equipment used during various manufacturing stages of the gyro assembly (see Drawing 639AS3180 or Drawing 639AS6502). The models or products in effect at date of issue of this standard are considered to be acceptable for use in manufacturing this assembly. However, any changes in the form of modifications may render the equipment unacceptable for use. Authorization to use the modified equipment may be obtained from the cognizant Government representative. Divided into two categories, the equipment and products are as follows.

4.2.1 AI Mechanical equipment.

4.2.1.1 AI Gisholt Dynamic Balancing Machine, Gisholt Model No. HS-1, Gilman Engineering and Manufacturing Company, 305 West Delavan Drive, Janesville, Wisconsin 53545. (Use Shop Accesory (SA) Tooling SA 2875049.)

4.2.1.2 AI Shrader Magnetizer, MHD4-5, Bux Shrader Company, 12080 South Western Avenue, Garden Grove, California 92641.

4.2.1.3 AI Precision Pivot Bearing (ABEC-7 or better), Miniature Precision Bearings, Incorporated, Precision Park, Keene, New Hampshire. Catalog No. S6A7B.

4.2.1.4 AI Federal Electro-Probe Electronic Test Indicator, Model 230P-3-R2, Federal Products Corporation, Providence, Rhode Island 02901.

4.2.2 AI Auxiliary items.

4.2.2.1 AI Ink, MArking Stencil, Opaque, in accordance with TT-I-1795, Type I.

4.2.2.2 AI Deltronic Tenth-Plug Gage Set, HI-Precision Grinding, Incorporated, Costa Mesa, California 92626.

4.2.2.3 AI Radio Frequency Laboratory Model 990 Magnetreater, RFL Industries, Powerville Road, Boonton, New Jersey 07005.

4.2.2.4 AI Cobehn Spray, Cobehn, Incorporated, 226 Passaic Avenue, Fairfield, New Jersey 07006. Cage Code 04882.

4.2.2.5 AI Hamilton MIcroliter Syringe, No. 710 with Cheney Adaptor (Hypodermic syringe with barrel graduated in microliters; No. 26 needle), Hamilton Company, P.O. Box 307, Whittier, California 90608.

4.2.2.6 AI Texwipe, Foamwipe, Foam Cubes, Texwipe Company, Hillsdale, New Jersey 07642. Cage Code 21994.

4.2.2.7 AI Primer, in accordance with MIL-S-22473 (Grade N, Form optional).

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4.2.2.8 AI Sealant, in accordance with MIL-S-22473 (Grade AA or AV).

4.2.2.9 AI Adhesive, in accordance with MIL-A-46050, Type I, Class 1.

4.2.2.10 AI Adhesive, Epoxy, in accordance with Drawing 639AS1487.

4.2.2.11 AI Adhesive Epoxy-Polyamide, in accordance with Drawing 639AS1486.

4.2.2.12 AI Tungsten Carbide Spherical Ball (0.2344 diameter) AFBMA Grade 10, Hoover Ball Division of Hoover Ball Bearing Company, Erwin Plant, Erwin, Tennessee 37650. Cage Code 29337.

4.2.2.13 AI Acetone in accordance with O-C-265, Reagent Grade.

4.2.2.14 AI Ethanol, Grade 1 in accordance with O-E-760, Class A or B.

4.2.2.15 AI Cement, in accordance with Drawing 639AS1611.

4.3 AI Solvent Warning. Post the following warning at each work station:

WARNING AI

Prolonged breathing of solvent vapors can be dangerous. Proper ventilation is necessary in any area where they are used. Acetone and ethanol are highly flammable liquids. Fire-hazard precautions and proper ventilation are recommended in any area where they are used. Treat all solvents with caution since prolonged breathing of vapors or skin contact is injurious.

4.3.1 M Solvent purity. Use only high-purity, reagent-grade solvents that have been filtered through a 0.5 micrometer filter in all processes specified herein.

4.4 Preassembly information.

4.4.1 AI SA tooling. This section provides information on SA fixtures of recommended SA equipment used in various phases of assembly or machining. Substitutes or innovations may be used if results prove satisfactory. Familiarity with SA drawings is useful, and each operator should be acquainted with the SA tool to the extent that they know when the tool is damaged or not capable of performing the operation correctly.

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adhesion and subsequent failure of a unit. Ensure that all mating surfaces are thoroughly dry, clean, and free from grease, oil, and other contaminants before application of any adhesive. The adhesive should wet the surface to form a satisfactory bond. Exercise discretion when applying adhesive so that contamination of functional surfaces does not occur.

CAUTION AI

Chemically blackened parts: Parts that are chemically blackened should not be subject to prolonged ultrasonic cleaning since such cleaning can be detrimental to the blackened areas.

4.5.1.1 MXM Mixing and curing adhesives. Mix and cure adhesives in accordance with manufacturer's instructions unless otherwise specified herein or on the appliable drawings. Accelerated curing temperature may be not greater than 90 degrees Celsius (°C). Use of colorant is optional.

4.5.1.2 AI Surface preparation. To check for proper surface preparation, a water-break-free test is recommended. This is a check of the surface tension of the surface to be joined. Distilled water applied with an eye dropper or equal may be used. If the surface is clean, the water will spread out and wet the surface and be free of breaks or water-break-free. If the surface is contaminated, the water will not spread or "wet" but will form droplets and have water breaks in areas where the water does not wet the surface. For a discussion of adhesive use and surface preparation, refer to the book "Adhesive Bonding Techniques and Applications," by Charles V. Cagle.

4.5.1.3 AI Sealant application. When applying sealant (see MIL-S-22473), use an appropriate-size eye dropper, graduated in microliters, so that drops of 1 microliter or 2 microliters can be applied accurately. Because the adhesive sets up quickly in a vacuum, a hypodermic-needle syringe is not recommended. The syringe depends on the vacuum principle for loading the barrel. A common sewing needle held in a pin vise may be used if it is used in a manner that ensures proper application.

4.5.1.4 AI Primer application. When applying primer (see MIL-S-22473, Grade N), a hypodermic-needle syringe is highly recommended. Use a No. 26 needle whose barrel is graduated in microliters so that drops of 1 microliter and 2 microliters can be applied accurately. After using, clean the needle and barrel with acetone (see 4.5.2). Clean and properly store both the needle and barrel when not in use.

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4.5.2 MXM Solvents. Cobehn, acetone, or ethanol applied with swabs as lint-free as possible are used for solvent cleaning. When using any solvent on cemented and assembled parts, exercise manual control in a manner that will not flood the spray-cleaned surfaces; such flooding will destroy uncured adhesive bonding. Since swabs are not completely lint-free, keep all swabs in clean containers until used. After using, discard the swabs in covered, disposal containers. Use only small quantities of solvent at a time. When using, ensure that the solvents are kept in small clean containers that yield clean solvents at all times. Do not use the same solvents or swabs more than once since this will result in contamination (see 4.3.1). Solvents and air used for solvent spray shall be filtered to remove particulate contamination greater than 0.5 micron.

4.5.3 MXM References. "Acetone swab," "ethanol swab," or "damp swab" mean that a dry swab or Foam Cube is dipped in the approporiate solvent in such a manner that, on application, the swab is damp but not dripping. This is accomplished by dipping a clean swab in the solvent and removing the excess by blotting on a folded Foamwipe or equal. Since bonding in many areas will be adversely affected, do not use dripping swabs during any assembly operation where adhesive damage may occur.

4.5.4 M Contamination prevention. Avoid unnecessary cleaning on mirrors, gimbal assembly, spin bearings, gimbal bearings, and the finished gyro assembly. Handle unmounted bearings in clean work stations as described in AD-874 when not in a protective container. Particular attention is drawn to the spin bearings which contain phenolic cages that are impregnated with lubricating oil to act as an oil reservoir for long storage. The bearings are grease-packed to provide a uniform film of lubricant over the working surfaces of the bearing. Cleaning with solvents will destroy both the grease-packing and the oil impregnation. Do not use a swab near enough to any bearing to cause contamination of the bearing. Relubrication of contaminated spin bearings by a qualified lubrication facility is required in the event of bearing contamination. For proper lubrication of gimbal bearings, refer to MIL-STD-2081. Do not handle bearings in any manner except with bearing handling tools or finger cots.

4.5.5 MXM Optical painted parts. Do not touch or handle optical painted surfaces unless obsolutely necessary, and never with bare fingers. Fingerprints and scuff marks cause detrimental effects on gyro performance. Ensure that optical elements are kept free of grease, oil, dust, lint, etc. During assembly procedures, exercise care and remove contamination carefully whn necessary.

4.5.5.1 AI Contamination. Low spectacular reflectance black surfaces depend in large measure on uniform unsoiled multifaceted surfaces. Contamination by body oils, other contaminants, and scuff marks due to

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CAUTION M

When installing balance screw, do not violate 0.088-inch minimum clearance from stud forward face to balance screw forward face (see Figure 2). This clearance is required to permit use of a pivot ball bearing for dynamic balancing phases. Do not cement screw at this time; adjustments may be required later (see 5.3.17).

5.3.11 MXM Fore and aft balance adjustment. Adjust the fore and aft balance of the gyro optical assembly to meet the requirements of MIL-G-85105. The test fixture used shall incorporate a simulated missile, including a head coil assembly, requisite electronics circuitry, and a rotating device. Perform the test with the gyro optical assembly mounted in the head coil assembly on the rotating device, and with the gyro rotating at regulated speed with the spin axis of the gyro and support post in a horizontal plane. Monitor the two current inputs to the servo amplifiers while the head coil assembly is rotated about the gyro spin axis at a rate of 4.0 ± 0.2 degrees per second. Compute the mass unbalance drift rate from the current input to the servo amplifiers. If the computed drift rate is greater than 0.025 degree per second, readjust the gravity balance in accordance with 5.3.10. Repeat the testing and readjustment alternately, as many times as necessary, to obtain a drift rate not greater than 0.025 degree per second in each input to the servo amplifier.

5.3.12 MXM Balance weight cleaning. Clean dynamic balance weights (Item 7, Figure 2) with acetone or Cobehn.

5.3.12.1 MXM Coating. Coat the outside spherical diameter of the mirror magnet with ink, marking stencil, opaque, in accordance with A-A-208, Type I, color black or blue starting at the north pole within 5 degrees proceeding clockwise, when looking into mirror face. Coat the spherical diameter as required to activate the photocell of the balance equipment (see 5.3.13.1).

5.3.13 M Dynamic balance adjustment. Perform dynamic balance adjustment as follows:

5.3.13.1 M Procedure. Set up gyro optical assembly in dynamic balancing machine, using fixtures (see Drawing SA 2875049 and Drawing SA 492499), and a Gisholt HS-1 balancing machine as adapted for gyro optical assembly. While balancing, use a gyro speed of 5,000 ± 50 revolutions per minute (rpm) for consistency during checking at various activities.

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5.3.13.1.1 MXM Calibration. Periodically calibrate the dynamic balancing machine as follows:

NOTE: MXM Calibration as outlined in 5.3.13.1.1.2.1 is performed once a month. Calibration of the 150 and 300 micro-ounce-inch points is performed once a week at a time agreeable to the Government representative or at any time the Government representative deems necessary to ensure quality balancing. Table I contains a list of required equipment. Also refer to Tables II and III. Tables IV and V show estimated values for production. This test is performed with a different standard production gyro for each calibration period. The use of master gyros for calibration is acceptable as a reference only.

TABLE I. MXM Required equipment for calibration.

Description
1. Standard production gyro (master gyro optional for reference).
2. Calibration weights as indicated in Table II.
3. 1/ Adhesive, MIL-A-46050, Type I, Class 1 or an equal fast-drying cement that permits quick application in minute amounts. Cement must be able to be removed easily upon completion of test.
4. Dynamic balancing machine with an amount meter calibrated as indicated in 5.3.13.1.1.1.
5. Fixture, Drawing SA 492499, Weight Orientation Fixture.

1/ MIL-A-46050 cement is quickly removed by using a small electrical soldering iron or electrical soldering pencil held briefly against weights applied to the gyro, since temperatures of approximately 170°C will rapidly weaken the bond.

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b. MXM When testing each weight in a given plane (magnet or sunshade), a check will simultaneously be made of the opposite plane to demonstrate that not greater than 10 percent crosstalk exists between planes. Select a weight which will produce an unbalance indication of not less than 50 percent of full scale on the meter.

c. MXM The angle indicator reading to be accurate within ± 10 degrees of the known angle at which the known weight was applied.

d. MXM Failure to meet the preceding test specifications requires that the balancing machine be readjusted and then retested to the satisfaction of the Government representative.

5.3.13.2 M Balancing. Hold gimbal in one fixed horizontal position during balancing. This requires a two-point support at stud centerhole and at gyro support post in accordance with 4.5.10.

CAUTION AI

Worn out pivot balls (0.2344 diameter) can cause erratic results during calibration, collimation, and dynamic balancing operations. Use several balls during monthly calibration to expose potential problems.

5.3.13.3 M Two-plane correction. Planes of correction are forward face of sunshade and aft side of mirror-magnet (see Drawing 639AS3180 or Drawing 639AS6503 for plane location). Cement weights (Item 7, Figure 2) in place with adhesive (see MIL-A-46050) or cement (see Drawing 639AS1611) after proper weight and angular location are determined. Push the weights firmly into place so as to contact both surfaces of the magnet (see 5.3.14). Tolerance in each plane is 150 micro-ounce inches maximum. Use fixture (see Drawing SA 492499) for proper location of correction weights. After drying, verify that the balance is within specification.

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CAUTION M

When moving weights or cleaning excess adhesive, thin filaments of adhesive may be produced which may contaminate assembly if care is not exercised.

CAUTION M

Any physical changes to gyro-optic spinning mass (such as phasing, focusing, and large optical corrections) require that gyro be rechecked through dynamic balancing to ensure that it still lies within specification.

5.3.13.3.1 MXM Adhesive and paint application. After completion of dynamic balance, apply adhesive, methacrylate (see Drawing 639AS1574) for a permanent bond and paint weights in forward plane with art brush or equal, using black optical lacquer in accordance with Drawing 2104343 or black ink in accordance with A-A-208, Type 1. Use extreme care to prevent damage to remainder of assembly.

5.3.14 M Verification. Verify that no rotating part touches any of the stationary parts.

5.3.15 M Optical scan diameter. Perform scan diameter measurement (optical balance). Verify that the gyro assembly (see Drawing 639AS3180 or Drawing 639AS6502), as installed in the seeker section (see Drawing 639AS3100), meets the scan diameter requirement of MIL-G-85105. If out of specification, adjust as follows.

5.3.15.1 MXM Cleaning. Clean optical balance weights (Item 8, Figure 2) with acetone or Cobehn.

5.3.15.1.1 M Gyro assembly adjustment. Using the proper amount of weight for optical balance, adjust gyro assembly by installing weights in sunshade so the requirements of 5.3.15 are met. Use cement technique as specified in 5.3.13.3 and paint weights as specified in 5.3.13.3.1. Observe CAUTION note of 5.3.13.3, except that after optical balance has been completed, the final dynamic specification for each plane is as follows:

Magnet plane - 155 micro-ounce inches maximum.

Sunshade plane - 200 micro-ounce inches maximum.

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